Berry extracts exert different antiproliferative effects against cervical and colon cancer cells grown in vitro.

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Polyphenol-rich berry extracts were screened for their antiproliferative effectiveness using human cervical cancer (HeLa) cells grown in microtiter plates. Rowan berry, raspberry, lingonberry, cloudberry, arctic bramble, and strawberry extracts were effective but blueberry, sea buckthorn, and pomegranate extracts were considerably less effective. The most effective extracts (strawberry > arctic bramble > cloudberry > lingonberry) gave EC 50 values in the range of 25-40 microg/(mL of phenols). These extracts were also effective against human colon cancer (CaCo-2) cells, which were generally more sensitive at low concentrations but conversely less sensitive at higher concentrations. The strawberry, cloudberry, arctic bramble, and the raspberry extracts share common polyphenol constituents, especially the ellagitannins, which have been shown to be effective antiproliferative agents. However, the components underlying the effectiveness of the lingonberry extracts are not known. The lingonberry extracts were fractionated into anthocyanin-rich and tannin-rich fractions by chromatography on Sephadex LH-20. The anthocyanin-rich fraction was considerably less effective than the original extract, whereas the antiproliferative activity was retained in the tannin-rich fraction. The polyphenolic composition of the lingonberry extract was assessed by liquid chromatography-mass spectrometry and was similar to previous reports. The tannin-rich fraction was almost entirely composed of procyanidins of linkage type A and B. Therefore, the antiproliferative activity of lingonberry was caused predominantly by procyanidins.